ELEC70090
EE9-CS1-3

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE/EIE PART IV: MEng and ACGI

DIGITAL CONTROL SYSTEMS

Tuesday, 13 May 2025, 10:00 AM

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : Y. Gao
Second Marker(s) : D. Angeli

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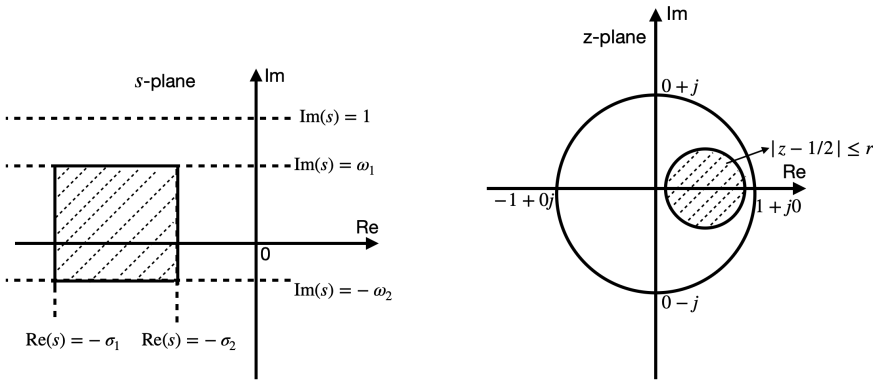
Information for candidates:

- $Z(\frac{1}{s}) = Z(1(kT)) = \frac{z}{z-1}$
- $Z(\frac{1}{s+a}) = Z(e^{-akT}) = \frac{z}{z-e^{-aT}}$
- $Z(\frac{\omega}{s^2+\omega^2}) = Z(\sin(\omega kT)) = \frac{z\sin(\omega T)}{z^2-2z\cos(\omega T)+1}$
- $Z(\frac{s}{s^2+\omega^2}) = Z(\cos(\omega kT)) = \frac{z(z-\cos(\omega T))}{z^2-2z\cos(\omega T)+1}$
- Transfer function of the zero-order hold: $H_0(s) = \frac{1-e^{-sT}}{s}$
- Definition of w-plane: $z = \frac{1+\frac{wT}{2}}{1-\frac{wT}{2}}$, $w = \frac{2}{T}\frac{z-1}{z+1}$
- Note that, for a given signal r (also indicated as $r(k)$, $r(kT)$ or $r(t)$), $R(z)$ denotes its Z-transform. All signals are assumed to be zero for negative times.
- Consider the characteristic equation is $1 + KF(z) = 1 + K\frac{N(z)}{D(z)}$. Suppose that $N(z)$ has roots z_i, $i = 1, \ldots, m$, $D(z)$ has roots p_i, $i = 1, \ldots, n$. The difference between n and m is the relative degree $r = n - m$.

  If $r > 0$ there are asymptotes of the root locus that intersect the real axis at $\sigma = \frac{\sum_{i=1}^{n} p_i - \sum_{i=1}^{m} z_i}{r}$, and radiate out with angles $\theta = \pm q\frac{\pi}{r}$, where $q = 1, 3, 5, \ldots$.

  Break-away or break-in points of the root locus exist where $N(z)D'(z) - N'(z)D(z) = 0$ ($'$ indicates the derivative).

1(A). Let the sampling time $T = 2$.



(a) s-plane (b) z-plane

Figure 1: s-plane and z-plane for question 1(A).

(i) Consider the shaded s-plane shown in Figure 1(a) where $\sigma_1, \sigma_2 > 0$, $\omega_1 > \omega_2 > 0$ and $\zeta > 0$ are given constants. Show that the shaded region lies in the primary strip. Sketch the corresponding z-plane region. [5 marks]

(iii) Consider the shaded z-plane shown in Figure 1(b) where $0 < r < 1/2$ is a constant. Sketch the corresponding w-plane region. [3 marks]

1(B). Consider the difference equation

$$x(k+2) + \alpha x(k+1) + \beta x(k) = 0$$

where α and β are given non-zero constants and $\alpha^2 - 4\beta > 0$.

(i) Determine the function $X(z)$. [2 marks]

(ii) No assumption on the initial values of $x(k)$ is made at this point. Compute the solution $x(k)$, in terms of $x(0)$ and $x(1)$. [4 marks]

(iii) Assume that the initial values $x(0)$ and $x(1)$ are arbitrarily finite reals. Find the condition on α and β such that the values $x(k)$ at $k = 2, 3, \ldots$ are bounded for any initial $x(0)$ and $x(1)$. Sketch the region in the α-β plane. [2 marks]

1(C). Consider the digital control system in Figure 2. Assume that the samplers are impulsive samplers.

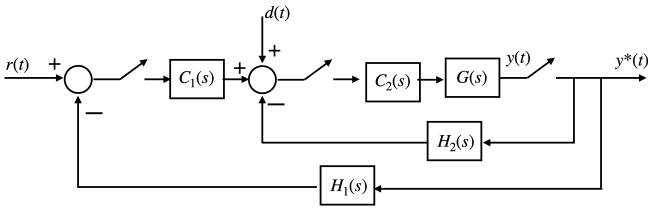


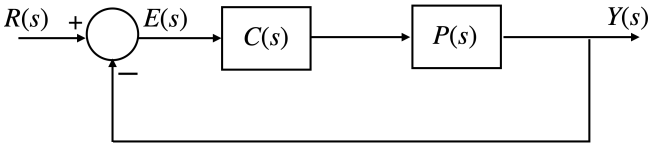
Figure 2: Diagram for question 1(C).

Determine the pulse transfer function between the input $R(z)$ and the output $Y(z)$ when $d(t) = 0$. And determine the pulse transfer function between the input $D(z)$ and the output $Y(z)$ when $r(t) = 0$. [4 marks]

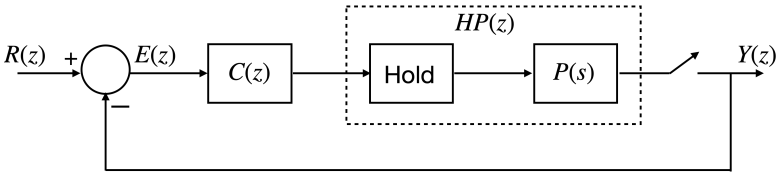
1(D). Consider a unity feedback discrete-time system with open-loop transfer function

$$G(z) = \frac{2}{z^3 + 2z^2 + z}.$$

Verify that the closed-loop system is not stable. [5 marks]



(a)



(b)

Figure 3: Block diagrams for question 2.

2(A). Consider a unity feedback system with open-loop continuous-time transfer function

$$P(s) = \frac{k}{s(s+1)}$$

interconnected, by means of a ZOH and a sampler, to the digital controller $C(z) = 1$.

(i) Compute the discrete-time equivalent transfer function. [4 marks]

(ii) Determine the exact set of values of k for which the closed-loop system is asymptotically stable. [5 marks]

2(B). Consider the continuous-time plant

$$P(s) = \frac{1}{s+a}$$

and the controller

$$C(s) = \frac{K}{s+b}$$

with $a > b > 0$, in a unity feedback configuration.

(i) Determine for which values of K, a, and b the closed-loop continuous-time system is asymptotically stable. [3 marks]

(ii) Compute the equivalent discrete-time model $HP(z)$ for the plant interconnected to the zero-order hold and the sampler. [4 marks]

(iii) Discretize the controller $C(s)$ using impulse response invariance method (call this controller $C_I(z)$). [2 marks]

(iv) Sketch the root locus of $C_I(z)HP(z)$. Determine for which values of K the closed-loop discrete-time system is asymptotically stable. [5 marks]

(iv) Compare and comment the ranges of K for the continuous-time system and the discrete-time system. [2 marks]

3(A). Consider a continuous-time linear system

$$\begin{cases} \dot{x}(t) = Fx(t) + Gu(t) \\ y(t) = Hx(t) \end{cases} \tag{1}$$

where

$$F = \begin{bmatrix} 0 & a \\ -a & 0 \end{bmatrix}, \; G = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \text{ and } H = \begin{bmatrix} 0 & 1 \end{bmatrix}.$$

Here a are non-zero constants and $a > 0$. Let $T > 0$ be the sampling time.

(i) Show that the system (1) is controllable and observable. [4 marks]

(ii) Discretize the continuous-time state-space equations (1). To this end, determine the matrices A, B and C of the representation

$$\begin{cases} x(k+1) = Ax(k) + Bu(k) \\ y(k) = Cx(k) \end{cases} \tag{2}$$

when the input $u(t)$ is sampled and fed to a zero-order hold. (*Hint: recall that* $e^{Ft} = \mathcal{L}^{-1}[(sI - F)^{-1}]$, *where* $\mathcal{L}^{-1}$ *indicates the inverse Laplace transform operator.*) [4 marks]

(iii) Show that if and only if the sampling time T is such that

$$aT \neq k\pi, \; k = \pm 1, \pm 2, \ldots$$

then

(iii.1) (A, B) is reachable; [2 marks]

(iii.2) (A, C) is observable; [2 marks]

(iii.3) the input-output transfer function of discrete-time system (2) is non-zero. [4 marks]

3(B). Consider the system

$$x(k+1) = Ax(k) + Bu(k)$$

and the discounted performance index

$$J_N = \gamma^N x^\top(N) S x(N) + \sum_{k=0}^{N-1} \gamma^k \left(x^\top(k) Q x(k) + u^\top(k) R u(k) \right)$$

with $0 < \gamma \leq 1$ is a constant, Q and S are positive definite or positive semidefinite matrices, and R is a positive definite matrix.

(i) Define the cost-to-go function. [3 marks]

(ii) Derive, using the dynamic programming, the optimal control $u(k)$ that minimizes J_N when the horizon N is finite. [6 marks]